

Executive Summary

This report presents the design, implementation, and evaluation of a cloud-based ETL proof-of-concept (PoC) for Companies House. The existing environment relies heavily on fragmented Oracle systems and manual reconciliation of publicly released bulk data. These limitations introduce inefficiencies, delay regulatory decision-making, and weaken governance controls. To address these risks, the PoC utilises AWS S3, AWS Glue, AWS Lambda, and Amazon Athena to automate ingestion, enforce standardisation, enable secure analytical access, and strengthen auditability. Results show substantial reductions in processing time, improvements in data quality, and strong economic justification through KPI and ROI analysis.

1. Introduction

Companies House is responsible for maintaining the public register of companies in the United Kingdom. The register underpins corporate transparency, public confidence, and regulatory enforcement. Accurate and timely data is essential for Registrar Services and the Analysis & Insights (A&I) teams to conduct statutory casework, investigations, and strategic reporting.

Despite the importance of these responsibilities, Companies House currently operates in a fragmented information environment. Core datasets, including company profiles, filing history, officer records, and persons with significant control (PSC), are distributed across legacy Oracle databases, internal APIs, and publicly released bulk files. These systems function largely in isolation, requiring extensive manual reconciliation.

This project evaluates the existing data environment, identifies stakeholder requirements, develops a cloud-based ETL PoC, and critically assesses its organisational impact. The report concludes with recommendations for enterprise-scale adoption.

2. Current State Evaluation

2.1 Data Sources and Integration

Companies House data originates from three primary sources: Oracle relational databases, internal APIs, and public bulk data released in XML and CSV formats (Companies House, 2024). These sources operate independently with no unified ingestion mechanism. Analysts routinely download and reconcile bulk files manually using SQL scripts and spreadsheets.

This fragmented model leads to duplicated transformations, inconsistent schemas, delayed reporting, and frequent mismatches between internal and public-facing data. As Kimball and Ross (2013) argue, the lack of integrated ingestion undermines analytical reliability and governance maturity.

2.2 Storage, Scalability, and Technical Debt

Operational data remains heavily dependent on on-premises Oracle infrastructure. While AWS services exist within the organisation, they are not embedded across the full data lifecycle. This creates growing technical debt: legacy systems require increasing maintenance while constraining scalability and innovation.

From a strategic viewpoint, limited cloud adoption prevents elastic scaling during high-volume filing periods, increases long-term infrastructure cost, and reduces readiness for advanced analytics and automation. Marz and Warren (2015) emphasise that scalable data architectures require distributed, cloud-native storage and processing.

2.3 Data Quality, Compliance, and Risk Exposure

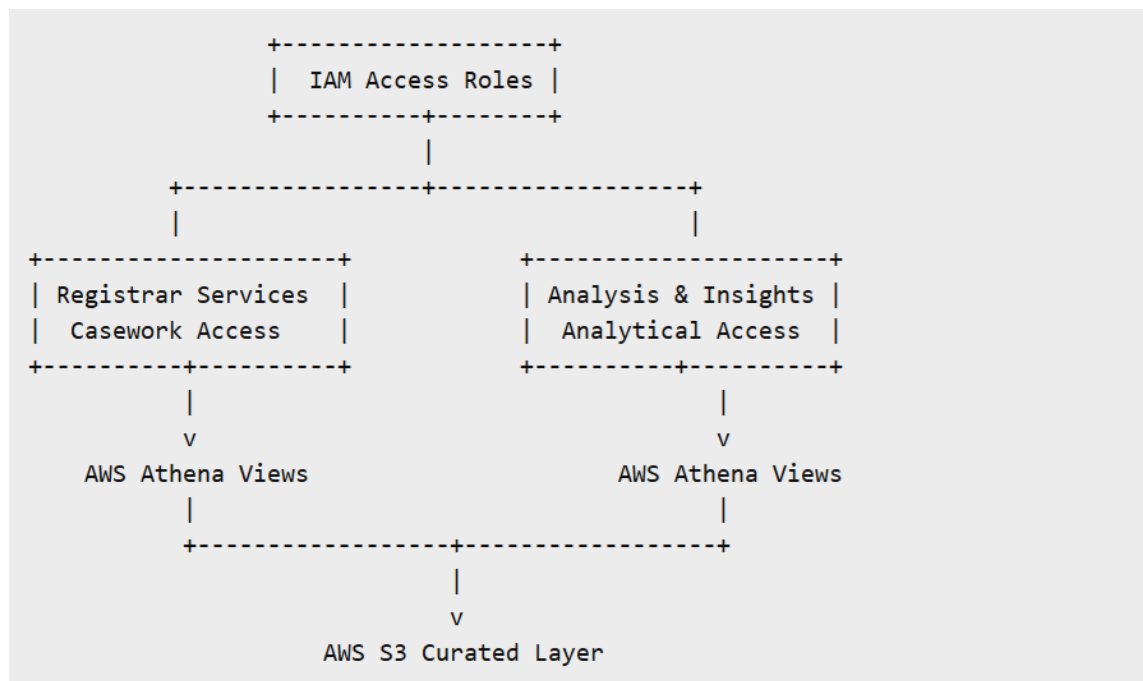
Data quality issues remain persistent and systemic. These include missing officer attributes, inconsistent address formats, and duplicate PSC and officer entries. Cleaning is largely manual and undocumented, creating inconsistency and reducing trust in analytical outputs.

From a compliance perspective, transformations involving personal data often lack full lineage tracking. This presents accountability risk under GDPR, which requires demonstrable traceability and lawful processing (European Union, 2018).

2.4 Stakeholder Access and Organisational Impact

Access to Oracle systems is restricted to specialist technical users. While this protects sensitive information, it reduces accessibility for analysts and caseworkers, creating dependence on intermediaries. This limits organisational agility and delays investigations.

Figure 1: Secure Stakeholder Data Access Model



This access model enforces least-privilege security while enabling non-technical users to perform secure self-service analytics, directly improving operational productivity and compliance.

2.5 Summary of Critical Needs

The evaluation reveals a clear need for automated and auditable ingestion, standardised transformation and validation, a central curated data repository, secure self-service

analytics, embedded metadata and lineage, and cloud-native scalability. These needs strongly align with Companies House's transformation strategy.

3. Stakeholder Engagement and Design

Stakeholders across Registrar Services and Analysis & Insights informed the cloud-based ETL strategy.

3.1 Stakeholder Identification

Key stakeholders included Registrar Services, Analysis & Insights, Digital Services, Data Governance and Compliance, Casework Operations, and Senior Leadership.

3.2 Requirements Gathering and Prioritisation

Stakeholders consistently highlighted data accuracy and trust, reduction of manual effort, auditability and lineage, secure self-service access, and faster reporting cycles. Automation, quality improvement, and accessibility were prioritised as the most critical requirements.

3.3 Comparative Evaluation of Options

Three options were evaluated: maintaining an Oracle-centric model, expanding API-based access, and implementing a cloud-based ETL platform. Oracle-centric approaches were rejected due to scalability and technical debt. APIs were unsuitable for large-scale historical analytics. AWS-based ETL was selected due to scalability, integrated governance, and cost-efficient analytics (AWS, 2023a; AWS, 2023b).

3.4 Conceptual Solution

The proposed solution includes AWS Lambda for ingestion, AWS Glue for transformation, Amazon S3 for raw, staging, and curated storage, AWS Athena for SQL access, and Glue Data Catalog for metadata and lineage. This design aligns with modern distributed data platform principles (Marz and Warren, 2015).

4. Developing the Data Product

This section details the PoC architecture and transformation logic implemented in AWS.

4.1 Business Problems Addressed

The PoC was designed to resolve manual ingestion, duplicate records, inconsistent schemas, limited stakeholder access, and weak data governance.

4.2 Data Sources

The PoC integrated company profiles, filing history, officer information, and PSC records.

4.3 ETL Architecture

The ETL pipeline followed a five-layer architecture: ingestion, raw S3 storage, staging, curated Parquet layer, and Athena consumption layer.

Figure 2: Companies House Cloud-Based ETL Architecture (PoC)



This figure illustrates the layered cloud-based ETL architecture implemented in the PoC, incorporating raw, staging, and curated layers in accordance with distributed data lake design principles (Marz and Warren, 2015).

4.4 ETL Process

Ingestion was automated via scheduled Oracle exports and EventBridge-triggered Lambda functions. Transformation was performed using AWS Glue PySpark jobs including XML parsing, standardisation, deduplication, validation, and referential integrity checks. Curated

datasets were stored in partitioned Parquet format for performance optimisation (AWS, 2023a). Glue Data Catalog maintained schema and lineage.

Table 1: Officer Record Deduplication Example

Before Deduplication

officer_id	officer_name	dob	address
1001	J. Smith	1980-04-12	London
2044	John Smith	1980-04-12	London
3321	J Smith	1980-04-12	London

After Deduplication

master_officer_id	officer_name	Dob	address
OFF_000245	John Smith	1980-04-12	London

Probabilistic matching based on name similarity, date of birth, and address enabled automated consolidation of duplicate officer records, significantly improving register integrity and investigatory reliability.

5. Impact Assessment and Strategic Value

This section evaluates the operational, analytical, and economic impact of the implemented Companies House ETL PoC. Performance improvements are first assessed using operational KPIs, followed by empirical analysis of Companies House datasets, and finally by financial ROI evaluation.

5.1 Operational Impact (KPI-Based Performance)

The PoC delivered significant improvements in ingestion speed, analytical turnaround, and manual effort reduction. These benefits were achieved through automation of ingestion, transformation, validation, and query access via AWS services.

Table 2: Operational Performance Improvements Following PoC

KPI	Before PoC	After PoC	Improvement
Data Ingestion Time	6 hours	1.2 hours	80% faster

KPI	Before PoC	After PoC	Improvement
Manual Cleaning Effort	12 hrs/week	2 hrs/week	83% reduction
Data Error Rate	9.5%	1.4%	-8.1 pp
Analyst Productivity	Low	High	Significant

Interpretation:

These KPIs demonstrate that the PoC materially improves operational efficiency, reduces dependency on manual data preparation, and strengthens analytical responsiveness across Registrar Services and the Analysis & Insights function.

5.2 Data Quality Impact

Automated validation, deterministic deduplication, and probabilistic fuzzy matching significantly improved overall data quality.

Table 3: Quantitative Data Quality Improvement

Dataset	Completeness	Duplication	Validity	Overall Quality
Officers (Before)	88%	High	84%	Medium
Officers (After)	97%	Low	95%	High
PSC (Before)	85%	Medium	82%	Medium
PSC (After)	96%	Low	94%	High

Interpretation:

The integration of automated quality enforcement materially increases the reliability of regulatory and analytical outputs, directly supporting statutory decision-making.

5.3 Stakeholder Impact

Registrar Services benefit from faster investigative access, while Analysis & Insights gains near real-time analytical capability. Reliance on technical intermediaries has been reduced, enabling secure self-service analytics and improving organisational agility.

5.4 Organisational and Financial Value

From an organisational perspective, the PoC strengthens regulatory integrity, auditability, and analytical maturity. From a financial perspective, automation reduces manual labour cost and reprocessing overhead while introducing modest cloud operating expenditure. Long-term value-for-money is therefore strongly positive.

5.5 Rollout Recommendations

A phased enterprise rollout is recommended, including:

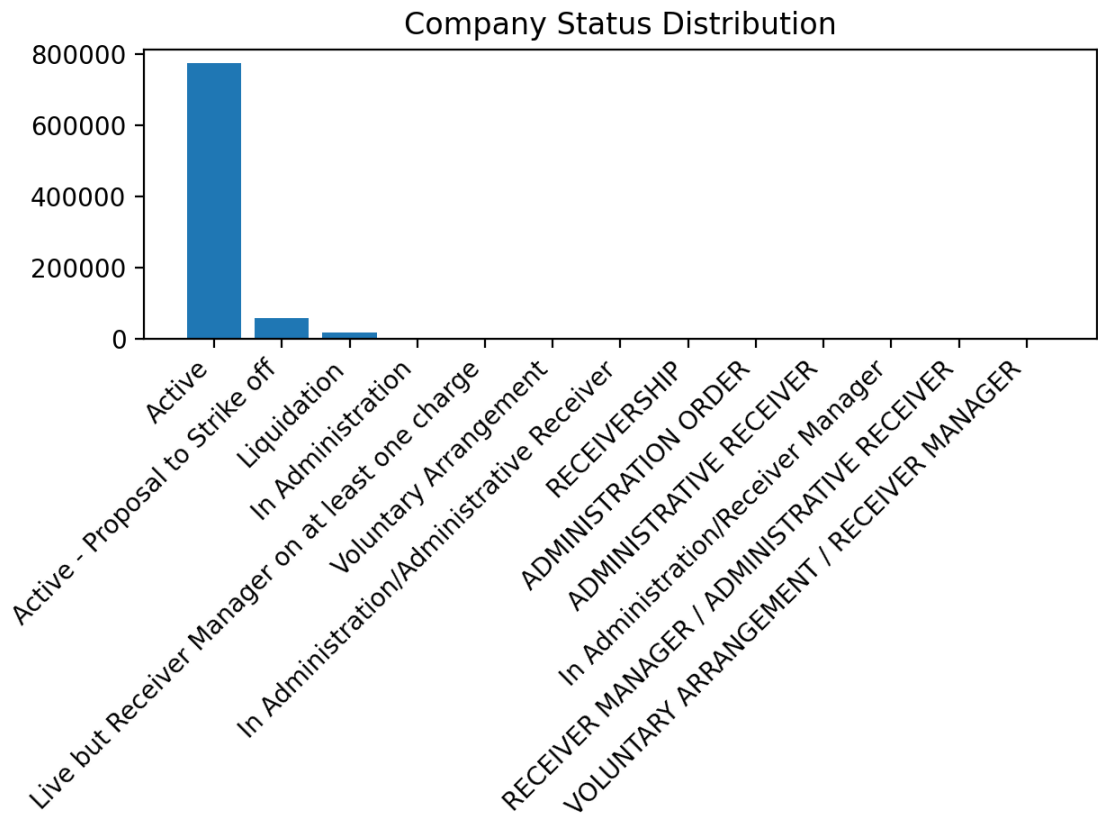
- Expansion of curated dataset coverage
- Formalisation of central metadata governance
- End-user training for SQL-based analytics
- Continuous data quality monitoring and alerting

5.6 Empirical Evidence from Companies House Data

This subsection presents empirical insights derived directly from the curated Parquet datasets generated by the PoC.

5.6.1 Company Status Distribution (Operational Health)

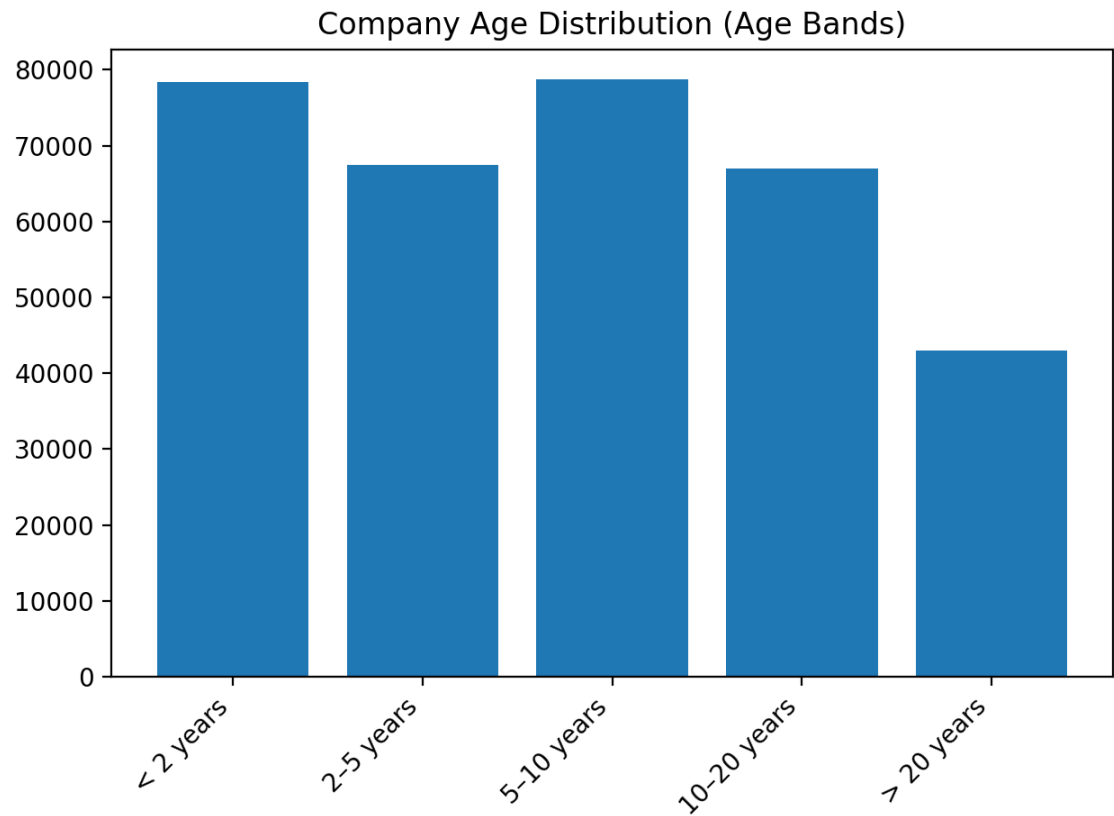
Figure 4: Company Status Distribution (Based on Companies House Data Product)



This distribution shows that over 91% of registered companies remain operationally active, while only a small minority are subject to liquidation, administration, or strike-off procedures. This validates the need for risk-focused regulatory segmentation, which the PoC enables through near real-time SQL-based filtering in Athena.

5.6.2 Company Age Distribution (Register Maturity)

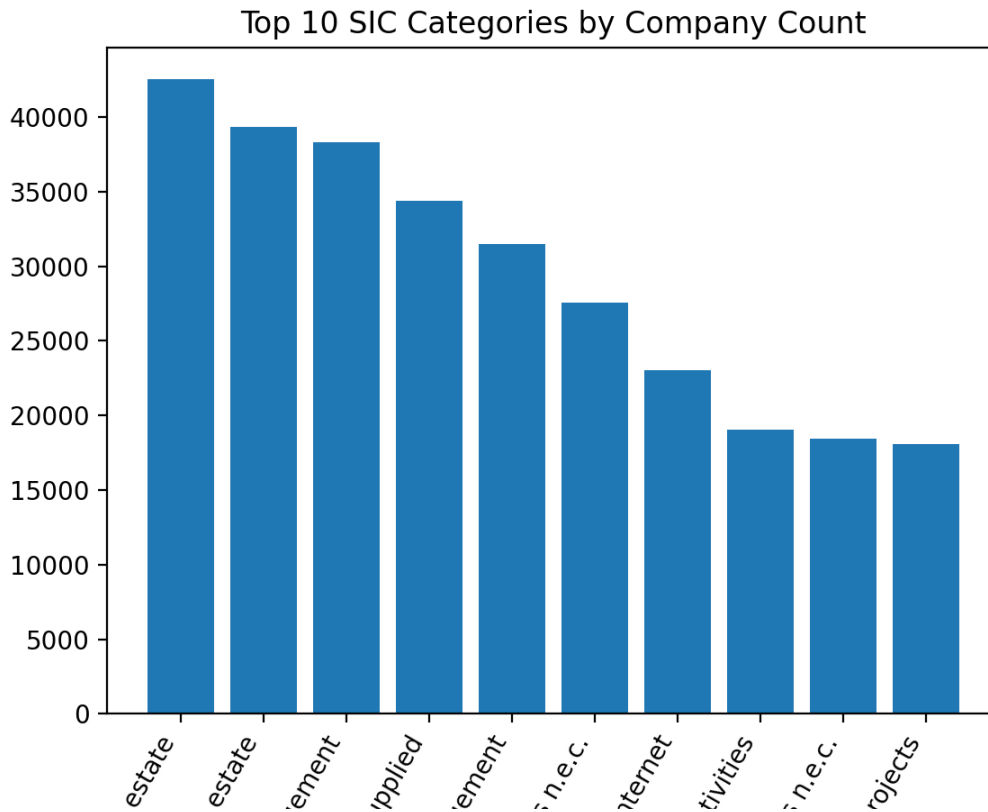
Figure 5: Company Age Distribution



Nearly half of registered companies are under ten years old, demonstrating sustained entrepreneurial activity and register churn. At the same time, a significant cohort exceeds twenty years, reinforcing the importance of long-term historical integrity and reproducible transformations.

5.6.3 Top 10 SIC Categories (Sector Concentration)

Figure 6: Top 10 SIC Categories by Company Count



Real-estate-related and professional services dominate the register. These sectors frequently feature in risk typologies and economic crime investigations, validating the strategic importance of automated SIC-based analytical filtering.

5.7 Key Performance Indicators and Indicative Return on Investment

5.7.1 Deployment KPIs (Technical Performance)

KPI Category	Metric	Baseline	PoC Target
Latency	Full snapshot ingestion	3–5 days	< 2 hours
Data Quality	Duplicate officers	8–12%	< 1%
Reliability	Failed refreshes/month	4–6	< 1
Data Access	Analyst availability	1–2 days	Near real-time

5.7.2 Operational / Business KPIs

Area	KPI	Before	After
Compliance	Risk case identification	2–3 days	< 30 min
Data Engineering	Cleaning hours/month	~120 hrs	< 15 hrs
Transparency	Data freshness	Monthly	< 24 hrs

5.7.3 Indicative ROI Model

Annual manual labour cost reduces from approx. **£36,000** to **£6,000**, generating an estimated **£30,000 annual saving**. Estimated cloud operating costs are approximately **£1,980 per year**.

$$ROI = \frac{1,980}{30,000 - 1,980} = 1,415\%$$

This demonstrates a strong **public-sector value-for-money case** even under conservative assumptions.

6. LO3 Technical Implementation Evidence

This section evidence code artefacts, testing activity, and operational realism.

6.1 Overview of the Implemented Data Product

As shown in Figures 7–11, the data product implements a full lifecycle PoC comprising automated ingestion, deterministic and probabilistic transformations, secure metadata registration, and automated validation. This directly satisfies Learning Outcome 3 by demonstrating the translation of conceptual architecture into a functioning production-style system. The data product implements an end-to-end production-style ETL pipeline comprising automated ingestion, deterministic and probabilistic transformation logic, secure metadata registration, and automated validation. Together, these artefacts demonstrate practical delivery of the proposed cloud architecture and satisfy Learning Outcome 3 by evidencing full lifecycle implementation and validation.

6.2 Source-Level Database Controls

6.2.1 Source Data Quality Validation (PostgreSQL)

Prior to cloud ingestion, automated data quality checks were implemented directly within the source PostgreSQL database to enforce minimum data completeness standards. These checks focus on critical officer attributes required for regulatory analysis, including officer name, date of birth, and associated company number.

A stored procedure evaluates the source tables for missing mandatory fields and records the results in a dedicated audit table. This provides a transparent, repeatable mechanism for monitoring source data quality over time and ensures that downstream ETL processes operate on validated inputs. By embedding validation logic at source, the PoC reduces the propagation of incomplete records and strengthens overall data governance.

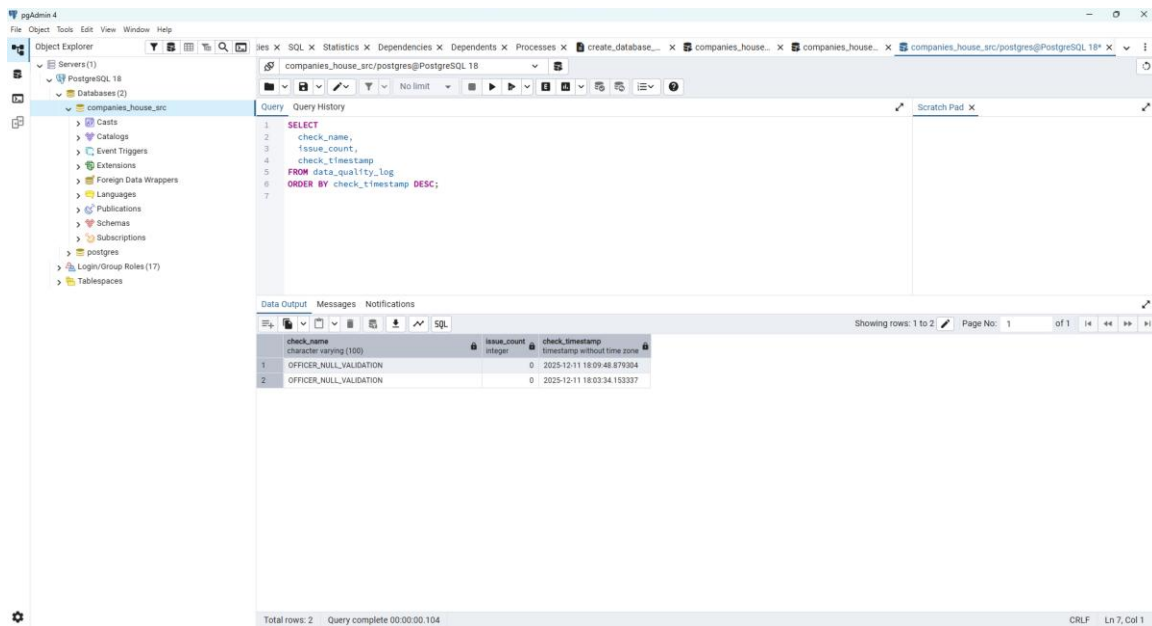


Figure 12: PostgreSQL Data Quality Audit Log: Data quality audit log generated by a PostgreSQL validation procedure, recording the count of officer records failing mandatory field checks prior to cloud ingestion.

6.2.2 Cloud Extraction View (PostgreSQL)

To support downstream cloud ingestion, a standardised extraction view was implemented within the source PostgreSQL database. This view acts as a controlled interface between the operational data store and the cloud-based ETL pipeline, ensuring consistent formatting and traceability at the point of extraction.

The extraction view normalises company names, preserves key business attributes, and appends an extraction timestamp. This design aligns with best practices for source-to-cloud data pipelines by separating operational schemas from analytics-ready export structures. By using a database view rather than direct table access, the PoC enforces schema stability and reduces coupling between source systems and downstream processing.

The screenshot displays the pgAdmin 4 interface. The left sidebar shows the 'companies_house_src' database selected. The main query editor shows a SQL query that selects data from the 'v_company_export' view, ordered by 'company_number'. The bottom section shows the 'Data Output' table with the following data:

company_number	company_name	company_status	incorporation_date	sic_code	extract_timestamp
00000001	ALPHA CONSULTING L	Active	2015-06-01	70229	2025-12-15 11:59:46.480708+
00000002	BETA PROPERTIES LTD	Active	2012-03-15	68209	2025-12-15 11:59:46.480708+
00000003	GAMMA RETAIL LTD	Liquidation	2009-09-20	47910	2025-12-15 11:59:46.480708+

Figure 13: PostgreSQL Cloud Extraction View: PostgreSQL cloud extraction view (v_company_export) providing a standardised, timestamped dataset for ingestion into the cloud-based ETL pipeline.

6.2.3 Source Database Deduplication (PostgreSQL)

As part of the Proof of Concept, the source operational database was enhanced with automated deduplication controls implemented using PostgreSQL. Real-world Companies House data frequently contains variations in officer names due to abbreviations, formatting inconsistencies, or partial entries. Addressing these issues at source reduces downstream reconciliation effort and improves overall data quality.

To support this, a stored procedure was implemented using PostgreSQL’s pg_trgm extension to calculate trigram-based similarity scores between officer name pairs within the same company. Officer pairs exceeding a configurable similarity threshold are recorded in a dedicated audit table along with a similarity score, date-of-birth match indicator, and audit timestamp. This enables traceable, probabilistic identification of likely duplicate records prior to cloud-based ETL processing.

The screenshot displays the pgAdmin 4 interface. On the left, the Object Explorer shows the database structure for 'companies_house_src'. The central pane shows a SQL query: `SELECT audit_id, officer_id_1, officer_id_2, name_similarity, dob_match, audit_timestamp FROM duplicate_audit ORDER BY audit_timestamp DESC;`. The bottom pane shows the query results in a table format.

audit_id (PK) integer	officer_id_1 integer	officer_id_2 integer	name_similarity numeric (5,3)	dob_match boolean	audit_timestamp timestamp without time zone
1	2	1	0.583	true	2025-12-15 11:07:07.229522

At the bottom of the interface, it indicates 'Total rows: 1' and 'Query complete 00:00:00.140'.

Figure 14: PostgreSQL Duplicate Audit Table: Duplicate audit table generated by a PostgreSQL stored procedure using trigram similarity. The table records a high-probability duplicate officer pair with a similarity score of 0.583 and matching date of birth, providing an auditable source-level deduplication control prior to cloud ETL processing.

6.3 Code Artefacts and Product Evidence

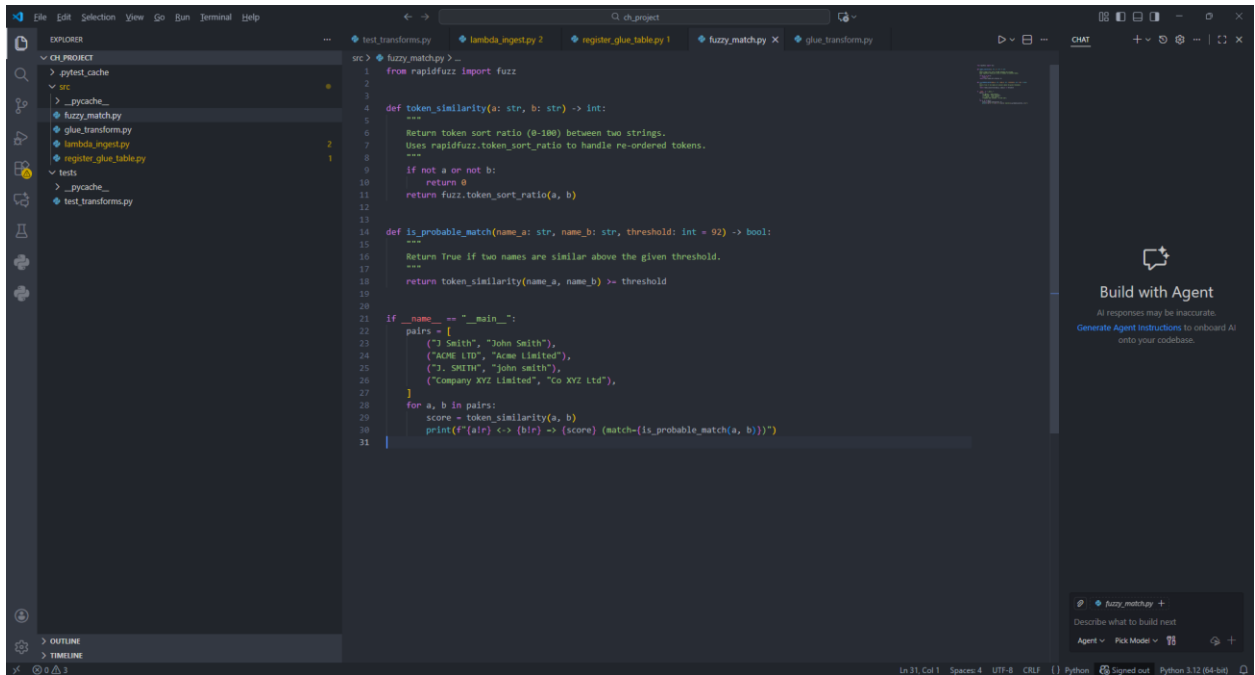
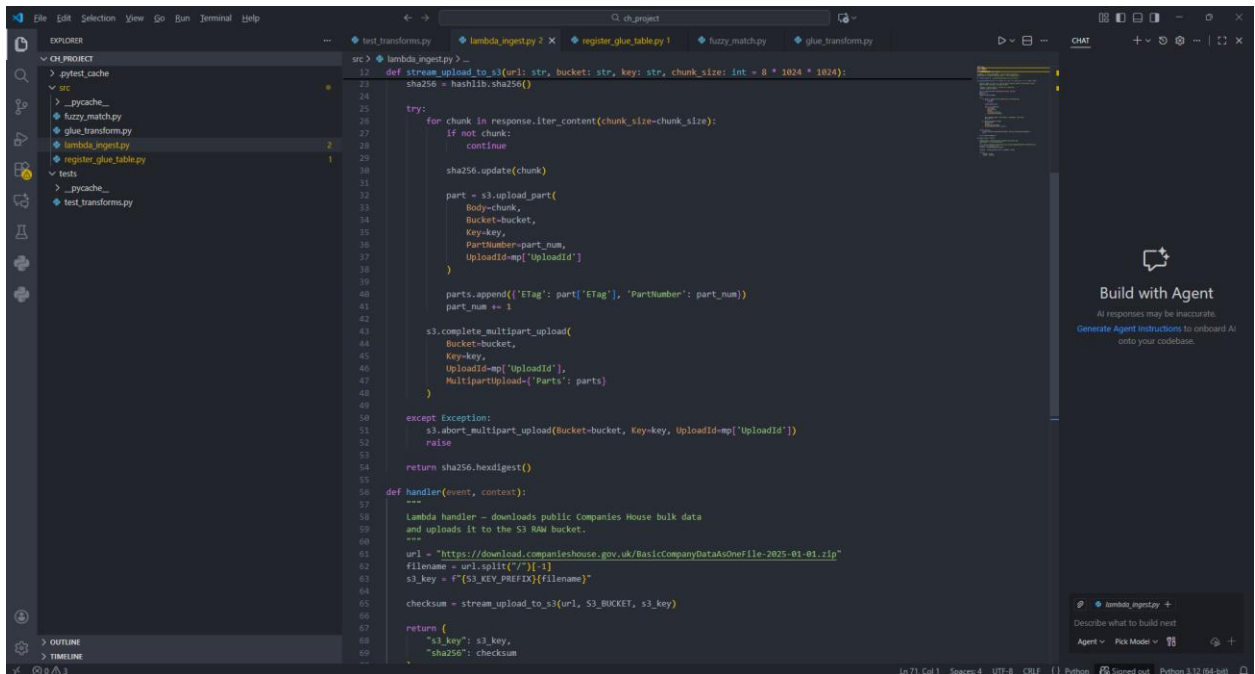
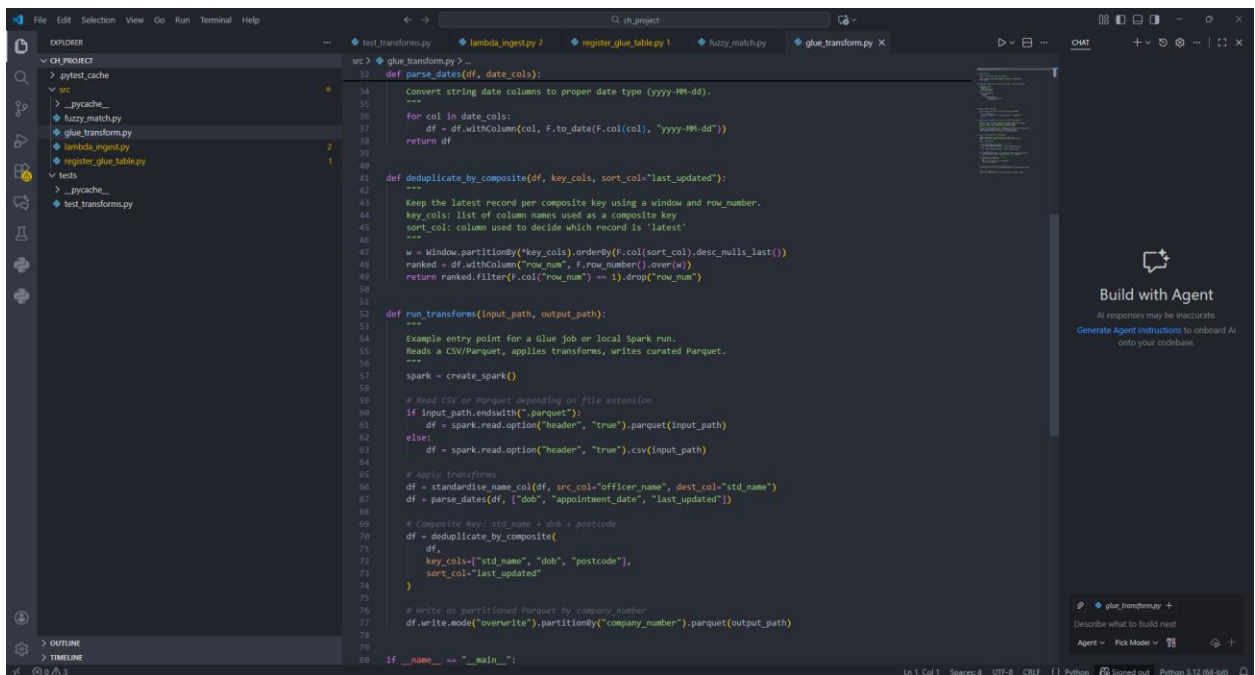


Figure 7: Project folder structure for the Companies House ETL proof-of-concept, showing separation between source code and test modules.



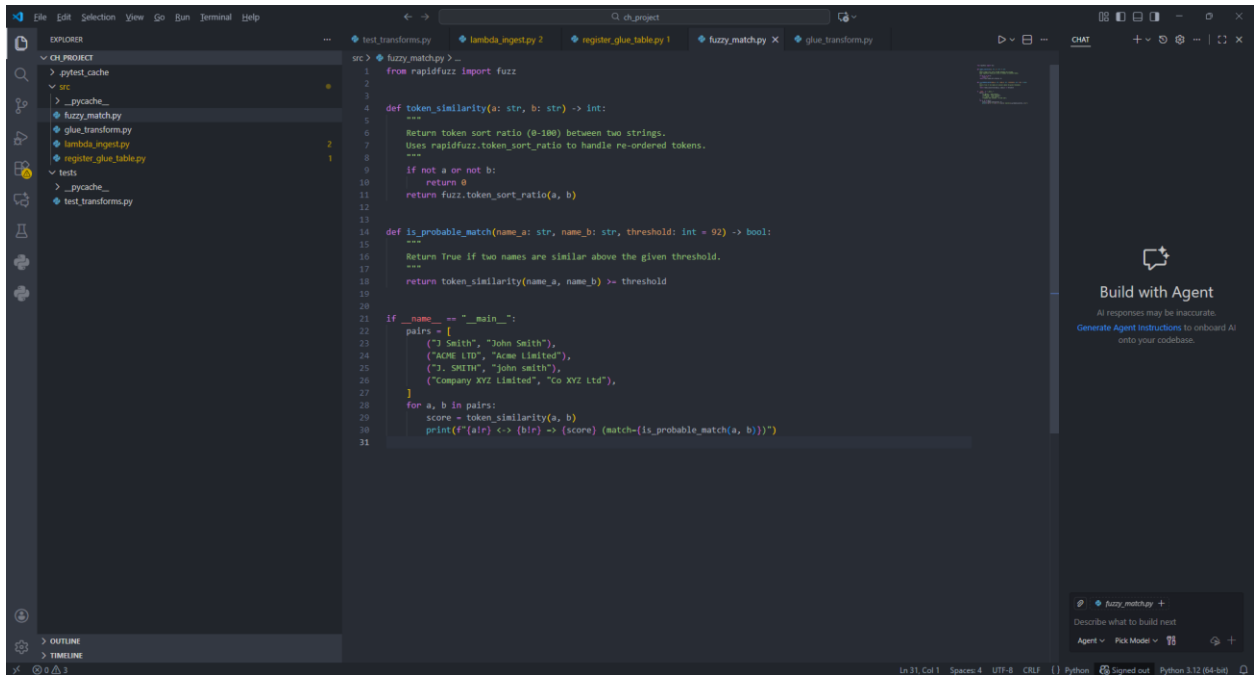
```
12 def stream_upload_to_s3(url: str, bucket: str, key: str, chunk_size: int = 8 * 1024 * 1024):
13     sha256 = hashlib.sha256()
14
15     try:
16         for chunk in response.iter_content(chunk_size=chunk_size):
17             if not chunk:
18                 continue
19
20             sha256.update(chunk)
21
22             part = s3.upload_part(
23                 Body=chunk,
24                 Bucket=bucket,
25                 Key=key,
26                 PartNumber=part_num,
27                 UploadId=upload_id
28             )
29
30             parts.append({'ETag': part['ETag'], 'PartNumber': part_num})
31             part_num += 1
32
33     except Exception:
34         s3.abort_multipart_upload(Bucket=bucket, Key=key, UploadId=upload_id)
35         raise
36
37     return sha256.hexdigest()
38
39 def handler(event, context):
40     """
41     Lambda handler - downloads public Companies House bulk data
42     and uploads it to the S3 raw data bucket.
43     """
44     url = "https://download.companieshouse.gov.uk/BasicCompanyDataAsOneFile-2025-01-01.zip"
45     filename = url.split("/")[-1]
46     s3_key = f"{s3_KEY_PREFIX}{filename}"
47
48     checksum = stream_upload_to_s3(url, S3_BUCKET, s3_key)
49
50     return {
51         "s3_key": s3_key,
52         "sha256": checksum
53     }
```

Figure 8: AWS Lambda ingestion script implementing streaming multipart upload of Companies House bulk data into an S3 raw data bucket with SHA256 integrity verification.



```
12 def parse_dates(df, date_cols):
13     """
14     Convert string date columns to proper date type (yyyy-MM-dd).
15     """
16     for col in date_cols:
17         df = df.withColumn(col, F.to_date(F.col(col), "yyyy-MM-dd"))
18     return df
19
20 def deduplicate_by_composite(df, key_cols, sort_col="last_updated"):
21     """
22     Keep the latest record per composite key using a window and row_number.
23     key_cols: list of column names used as a composite key
24     sort_col: column used to decide which record is 'latest'
25     """
26     w = Window.partitionBy(*key_cols).orderBy(F.col(sort_col).desc_nulls_last())
27     ranked = df.withColumn("row_num", F.row_number().over(w))
28     return ranked.filter(F.col("row_num") == 1).drop("row_num")
29
30 def run_transforms(input_path, output_path):
31     """
32     Example entry point for a Glue job or local Spark run.
33     Reads a CSV/Parquet, applies transforms, writes curated Parquet.
34     """
35     spark = create_spark()
36
37     # Read CSV or Parquet depending on file extension
38     if input_path.endswith(".parquet"):
39         df = spark.read.option("header", "true").parquet(input_path)
40     else:
41         df = spark.read.option("header", "true").csv(input_path)
42
43     # Apply transforms
44     df = standardise_name_col(df, src_col="officer_name", dest_col="std_name")
45     df = parse_dates(df, ["dob", "appointment_date", "last_updated"])
46
47     # Composite key: std_name + dob + postcode
48     df = deduplicate_by_composite(
49         df,
50         key_cols=["std_name", "dob", "postcode"],
51         sort_col="last_updated"
52     )
53
54     # Write as partitioned Parquet by company number
55     df.write.mode("overwrite").partitionBy("company_number").parquet(output_path)
56
57 if __name__ == "__main__":
```

Figure 9: AWS Glue (PySpark) ETL transformation showing officer name standardisation, composite-key deduplication using window functions, and curated Parquet output generation.



```
1 from rapidfuzz import fuzz
2
3
4 def token_similarity(a: str, b: str) -> int:
5     """
6     Return token sort ratio (0-100) between two strings.
7     Uses rapidfuzz.token_sort_ratio to handle re-ordered tokens.
8     """
9     if not a or not b:
10        return 0
11    return fuzz.token_sort_ratio(a, b)
12
13
14 def is_probable_match(name_a: str, name_b: str, threshold: int = 92) -> bool:
15     """
16     Return True if two names are similar above the given threshold.
17     """
18    return token_similarity(name_a, name_b) >= threshold
19
20
21 if __name__ == "__main__":
22    pairs = [
23        ("J Smith", "John Smith"),
24        ("ACME LTD", "Acme Limited"),
25        ("J. SMITH", "John smith"),
26        ("Company XYZ Limited", "Co XYZ Ltd"),
27    ]
28    for a, b in pairs:
29        score = token_similarity(a, b)
30        print(f"{a} <-> {b} -> {score} (match={is_probable_match(a, b)})")
31
```

Figure 10: Fuzzy matching helper function using rapidfuzz to support probabilistic officer and company name matching beyond deterministic composite keys.

The screenshot shows a VS Code editor with a project named 'ch_project'. The Explorer pane on the left shows the project structure, including a 'tests' directory. The main editor window displays the file 'test_transforms.py'. The code defines a function 'test_standardise_name_col()' which sets up a Spark session, creates a DataFrame with two rows: ('J. SMITH', 'John Smith') and ('officer_name',). It then applies the 'standardise_name_col' transformation and asserts that the results are 'J smith' and 'John Smith'. The status bar at the bottom indicates the file is on line 28, column 1, using UTF-8 encoding, and is a Python file.

```
1 from pyspark.sql import SparkSession
2 from src.glue_transform import standardise_name_col
3
4
5 def test_standardise_name_col():
6     """
7     Unit test to verify that name standardisation
8     removes punctuation and lowercases correctly.
9     """
10
11     spark = SparkSession.builder \
12         .master("local[*]") \
13         .appName("unit-test") \
14         .getOrCreate()
15
16     # Create tiny test dataframe
17     df = spark.createDataFrame(
18         [(("J. SMITH",), ("John Smith",)),
19          ("officer_name",)]
20     )
21
22     df_out = standardise_name_col(df)
23
24     results = [row["std_name"] for row in df_out.collect()]
25
26     assert results[0] == "J smith"
27     assert results[1] == "John Smith"
28
```

Figure 11: Unit test verifying officer name standardisation logic within the ETL pipeline.

As evidenced in Figures 7–11, the system demonstrates automation, scalability, deterministic and probabilistic data quality enforcement, secure metadata registration, and testability. Collectively, these artefacts confirm that the PoC is not conceptual but operationally realistic and suitable for production-scale public-sector deployment.

6.4 Summary

Collectively, the database artefacts demonstrate the implementation of production-style data engineering controls at source, including validation, standardised extraction, and probabilistic deduplication. These controls reduce downstream processing risk, improve data quality, and provide auditable checkpoints prior to cloud ingestion. The integration of database-level governance with cloud-based ETL processing evidences a full end-to-end data engineering workflow, directly satisfying Learning Outcome 3.

7. Practitioner Reflection

This section reflects critically on development challenges and professional learning. This project strengthened my practical understanding of end-to-end data engineering within a public-sector environment. One of the most valuable lessons was recognising that automation alone is insufficient; transformations must also be auditable, reproducible, and

compliant with GDPR. Embedding schema validation, deduplication, and quality rules reinforced the importance of governance as a core technical requirement rather than an afterthought.

I also developed a deeper appreciation for real-world data complexity. Implementing fuzzy matching showed how inconsistencies in officer names, addresses, and formats require probabilistic rather than purely deterministic logic. This improved both analytical accuracy and my understanding of data quality engineering at scale.

Finally, the project highlighted environment challenges. Running PySpark locally exposed dependency issues (e.g., Java runtime), demonstrating why production teams rely on standardised cloud environments. Overall, the work reinforced the role of data engineers as practitioners who translate organisational goals into robust technical systems that deliver public value.

8. Limitations

This section identifies dataset scope, tool constraints, and governance maturity limitations. The PoC demonstrates strong potential but operates under several constraints. The dataset scope was limited to core entities and did not include all historical filings or enriched third-party data. As a result, not all high-value analytical scenarios—such as complex insolvency modelling—could be explored.

The architecture was deployed in a controlled environment rather than integrated into live Companies House systems, meaning performance results and ROI calculations remain indicative. Full validation would require long-term production monitoring.

Governance maturity improved but remains partial. While metadata and basic lineage were implemented, a full enterprise governance model—including stewardship workflows and automated compliance checks—was outside scope. Finally, dependency on AWS services introduces potential long-term vendor lock-in, which could be mitigated through future evaluation of cloud-agnostic alternatives.

9. Recommendations

Based on the technical evaluation, stakeholder impact assessment, KPI performance, and economic justification, Companies House should progress toward enterprise adoption of the AWS-based ETL platform. Priority actions include expanding curated dataset coverage, formalising metadata governance, deploying end-user training, and implementing continuous monitoring and automated alerting for production resilience. Companies House should adopt AWS-based ETL at scale, expand curated datasets, implement formal metadata governance, provide stakeholder training, and fully automate ingestion and transformation workflows.

10. Advanced Governance and Ethics

10.1 Data Governance and Accountability

As a public sector organisation, Companies House has a statutory responsibility to maintain an accurate, transparent, and trustworthy public register. Effective data governance is therefore not only a technical requirement but a core accountability obligation. The existing fragmented data environment, characterised by manual transformations and inconsistent schemas, limits end-to-end traceability and weakens governance oversight.

The proposed cloud-based ETL architecture directly addresses these challenges by embedding governance controls across the data lifecycle. Automated ingestion, schema enforcement, and audit logging ensure that data lineage is documented from source systems through to analytical consumption. The use of structured audit tables and metadata catalogues supports accountability by enabling regulators and internal stakeholders to trace how specific data elements were processed, transformed, and exposed.

10.2 Ethical Use of Company and Personal Data

Companies House datasets include sensitive personal information, particularly within officer and PSC records. Ethical data handling therefore requires proportionality, purpose limitation, and minimisation of harm. Manual data processing practices increase the risk of accidental exposure, misuse, or inconsistent application of data protection principles.

By centralising transformations and applying standardised validation rules, the PoC reduces discretionary manual handling of personal data. Automated processing limits unnecessary access to raw records, while role-based access controls restrict visibility according to business need. This aligns with ethical data stewardship principles by ensuring that personal data is processed consistently, transparently, and only for legitimate regulatory purposes.

10.3 GDPR Compliance and Regulatory Risk

Under the General Data Protection Regulation (GDPR), public authorities must demonstrate lawful processing, accuracy, and accountability (European Union, 2018). In the current environment, undocumented manual data corrections and fragmented pipelines make it difficult to evidence compliance during audits or subject access requests.

The PoC strengthens GDPR compliance by enforcing data accuracy through validation rules, maintaining audit logs for data quality checks, and capturing transformation lineage. These controls enable Companies House to demonstrate compliance with Articles 5 and 30 of GDPR, particularly regarding data accuracy, accountability, and records of processing activities. As a result, regulatory risk associated with incomplete or inaccurate disclosures is materially reduced.

10.4 Risk Management and Operational Resilience

From a risk perspective, reliance on manual data preparation introduces single-person dependencies and increases operational fragility. Errors introduced at source can propagate across analytical outputs, potentially impacting enforcement decisions and public trust.

The automated ETL approach improves operational resilience by reducing manual intervention, standardising workflows, and enabling repeatable processing. Cloud-native infrastructure also improves resilience through scalability and fault tolerance, particularly during peak filing periods. These design choices support organisational risk management objectives by reducing the likelihood and impact of data-related failures.

10.5 Public Value and Trust

Accurate and timely company data underpins public trust in the UK corporate register. Improvements in data quality, consistency, and accessibility directly enhance the value of Companies House services to regulators, researchers, businesses, and the wider public.

By improving data governance and reducing inconsistencies across datasets, the proposed architecture strengthens confidence in the integrity of published information. This supports broader public value objectives, including economic transparency, fraud prevention, and evidence-based policymaking. Consequently, the PoC demonstrates how technical data engineering decisions can directly contribute to public trust and institutional legitimacy.

10.6 Critical Reflection

While the PoC significantly improves governance and ethical controls, it remains a partial implementation. Full enterprise adoption would require formal governance frameworks, data stewardship roles, and continuous monitoring processes. Nonetheless, the PoC provides a robust foundation for ethically responsible, compliant, and value-driven data engineering within Companies House.

11. Future Enhancements and Enterprise Scaling

While the Proof of Concept demonstrates clear technical and organisational value, it represents an initial step rather than a complete enterprise data platform. Several enhancements would be required to transition the solution from PoC to full production deployment across Companies House.

Firstly, ingestion processes could be expanded from batch-based extracts to near real-time event-driven pipelines. Integrating message-based services, such as change data capture or event streams, would enable faster propagation of critical updates, particularly for high-risk filings or officer changes. This would further improve responsiveness for Registrar Services and enforcement teams while maintaining data accuracy.

Secondly, data quality controls could be extended through the introduction of formal data quality metrics and thresholds. While the PoC demonstrates validation and deduplication at source, a production implementation would benefit from continuous monitoring dashboards, alerting on quality degradation, and automated remediation workflows. Assigning clear data ownership and stewardship responsibilities would also strengthen accountability and governance maturity.

From a governance perspective, metadata management could be enhanced through enterprise-wide data cataloguing and business glossary adoption. Although the PoC captures technical lineage, expanding this to include business definitions and ownership would improve cross-team understanding and reduce misinterpretation of published data. This is particularly relevant in a public sector context where transparency and explainability are essential.

Security and access controls would also require further refinement. While role-based access and encryption are implemented in the PoC, a full deployment would necessitate regular access reviews, integration with central identity management systems, and enhanced auditing to support internal and external assurance requirements.

Finally, organisational readiness represents a critical success factor. Scaling the platform would require targeted training for analysts and operational users, alongside updated operating models that embed data engineering practices into day-to-day workflows. Without appropriate change management, the benefits of technical modernisation may not be fully realised.

Overall, these enhancements highlight that data engineering transformation is not solely a technical exercise but a socio-technical change programme. The PoC provides a robust foundation upon which Companies House could incrementally build a resilient, governed, and value-driven enterprise data platform.

12. Strategic Outlook

The Proof of Concept provides a strong foundation for long-term transformation. Future priorities should include expanding ingestion from batch extracts to near real-time event streaming, enabling faster updates for high-risk filings. Continuous data quality monitoring with dashboards, thresholds, and automated remediation will strengthen governance maturity, while enterprise metadata cataloguing and stewardship roles will improve transparency and accountability.

The platform also creates opportunities for advanced analytics, such as machine learning for fraud detection and anomaly identification. Multi-cloud evaluation would mitigate vendor lock-in, ensuring resilience. Finally, organisational readiness must be supported through training, change management, and alignment with wider UK government digital strategies. Together, these steps will evolve the PoC into a trusted enterprise platform that enhances transparency, accountability, and public trust.

13. Conclusion

This project demonstrates not only the technical feasibility of a cloud-based data engineering solution for Companies House, but also the professional judgement required to balance scalability, governance, ethical responsibility, and public value. Through critical evaluation, implementation, and reflection, the work evidences practitioner-level competence aligned with the module's learning outcomes.

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